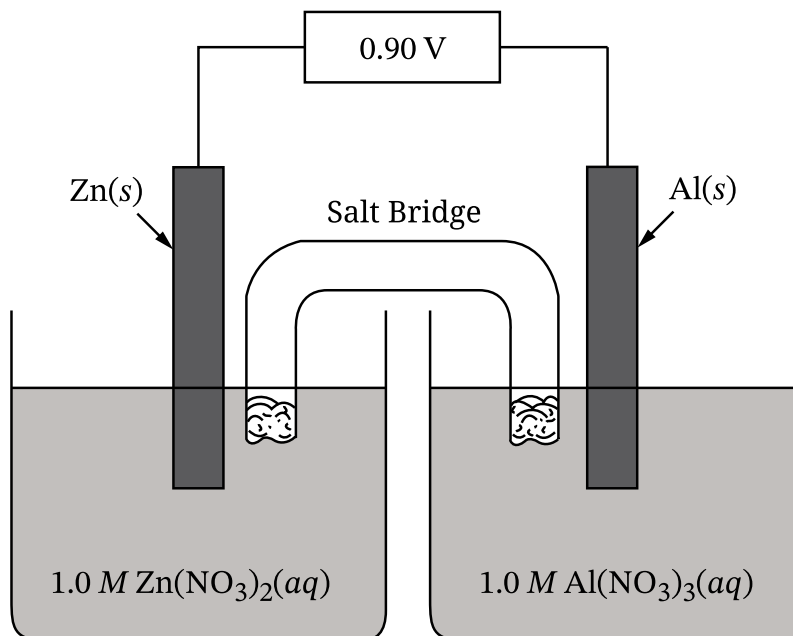


6. A scientist constructs a galvanic cell as shown in the diagram. As the cell operates, the $\text{Zn}(s)$ electrode increases in mass and the $\text{Al}(s)$ electrode decreases in mass. A data table with the standard reduction potentials for the substances follows the diagram.



Half-Reaction	E° (V)
$\text{Zn}^{2+}(aq) + 2e^- \rightarrow \text{Zn}(s)$	-0.76
$\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$	-1.66

- A. Write the half-reaction for the oxidation that occurs at the anode.
- B. Write the balanced net ionic equation for the overall reaction that occurs in the galvanic cell.
- C. Initially, each electrode has a mass of 50.0 g. The cell is allowed to run for a period of time and is then stopped. Which electrode's mass changed the most? Justify your answer with a calculation.

Reduction Half-Reaction	E° (V)
$\text{Au}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Au}(\text{s})$	+1.50
$\text{Zn}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Zn}(\text{s})$	−0.76
$\text{Mn}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Mn}(\text{s})$	−1.19
$\text{Al}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Al}(\text{s})$	−1.66
$\text{Be}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Be}(\text{s})$	−1.85

- D. The standard Zn/Al cell has a value of E_{cell}° equal to 0.90 V. The scientist needs a galvanic cell that produces a greater voltage. The scientist has access to the chemical systems in the table. If the scientist uses the Zn half-cell and one of the other options from the table, what is the MAXIMUM voltage that could be generated at standard conditions?